

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED  
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

|                   |                    |                                                                                              |
|-------------------|--------------------|----------------------------------------------------------------------------------------------|
| <b>Sec No- 12</b> | <b>Team No- 10</b> | <b>Project Title- AI-Based Network Intrusion Detection and Threat Intelligence Dashboard</b> |
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**1. Abstract**

The AI-Based Network Intrusion Detection and Threat Intelligence Dashboard is a database-driven cybersecurity system designed to identify, analyze, store, and visualize potential network security threats. With the rapid growth of digital networks, organizations generate large volumes of network traffic, making it difficult to manually detect malicious activities and security breaches. The proposed system addresses this challenge by integrating Artificial Intelligence (AI), Machine Learning (ML), Database Management Systems (DBMS), and threat intelligence into a unified platform. The system collects and analyzes network traffic and security-related data to identify abnormal patterns and potential intrusions such as unauthorized access, suspicious activities, and malicious network behavior. Machine learning techniques are used to classify network activities as normal or potentially harmful. The detected threats, including source and destination IP addresses, attack types, timestamps, severity levels, and alert details, are securely stored in a centralized relational database.

The DBMS component provides efficient data organization, storage, retrieval, and management using concepts such as primary keys, foreign keys, normalization, relationships, SQL queries, indexing, and views. The dashboard presents the collected information through interactive tables, statistics, and visual representations, enabling administrators to monitor network activity and identify security trends effectively. Historical threat records can also be retrieved for further analysis and reporting. The proposed system aims to improve intrusion detection efficiency, reduce manual monitoring, provide faster access to threat information, and support informed security decisions. By combining AI-based analysis with robust database management and threat intelligence, the system provides a scalable and efficient solution for network security monitoring and threat detection.

**2. Team Details**

| <b>Roll Number(s)</b> | <b>Name of the Student</b> | <b>Signature</b> |
|-----------------------|----------------------------|------------------|
| 2520090040            | P. Srivalli                | srivalli         |
| 2520090041            | V. Pavani                  | Pavani           |
| 2520090143            | A. Kavya Sree              | Kavya sree       |

### 3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

| Category                 | Framework / Tool      | Purpose                                                 |
|--------------------------|-----------------------|---------------------------------------------------------|
| Programming Language     | Python                | Implementing AI/ML models and backend logic             |
| AI/ML Libraries          | Scikit-learn          | Intrusion detection and Classification                  |
| Data Processing          | Pandas, NumPy         | Data cleaning, preprocessing, and analysis              |
| Database                 | MySQL                 | Storing network traffic, alerts, threats, and user data |
| Backend Framework        | Flask                 | Connecting the AI model, database, and dashboard        |
| Frontend                 | HTML, CSS, JavaScript | Designing the dashboard interface                       |
| Visualization            | Chart.js / Matplotlib | Displaying threat statistics and network activity       |
| Development Platform     | Visual Studio Code    | Project development and coding                          |
| Database Management Tool | MySQL Workbench       | Database design, SQL queries, and administration        |
| Version Control          | Git / GitHub          | Source-code management and collaboration                |
| Operating Platform       | Windows / Linux       | Development and execution environment                   |

Guide Name and Signature (remarks if any) : Dr. Khaja shareef